

RAFA

Regime-Aware Fiber Analysis

Quantifying fibre orientation and morphometry in composites
from 3D X-ray computed tomography

Conference presentation

Why fibre orientation matters

- Composite stiffness and strength are set by fibre orientation.
- Used in load-bearing aerospace, defence, and energy structures.
- X-ray CT images the 3D fibre architecture non-destructively.
- Challenge: reliable orientation across very different resolutions.

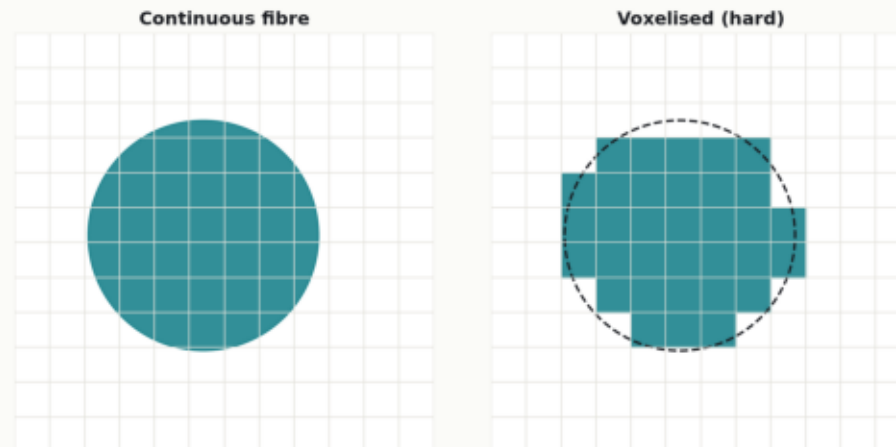
One idea: let the resolution pick the algorithm

- A single ratio drives everything: $r = \text{voxel spacing} / \text{fibre diameter}$.
- Resolved fibres are traced individually, one by one.
- Unresolved fibres are described by orientation-tensor statistics.
- RAFA selects the regime automatically from r .



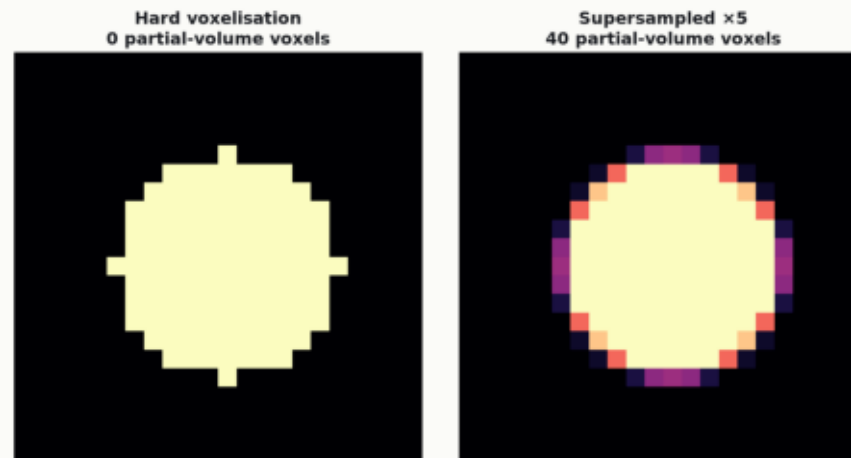
Voxelisation: geometry on a discrete grid

- Real CT and phantoms live on a voxel grid with known spacing (μm).
- That spacing is the fundamental length scale of the analysis.
- Anisotropic scans are resampled to isotropic voxels first.
- The voxel-to-fibre ratio r determines the regime.

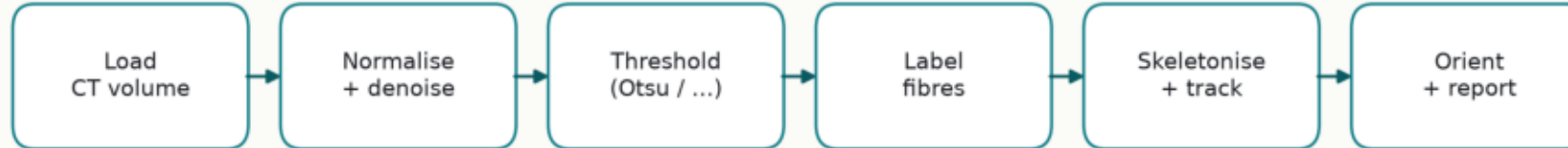


Supersampling: realistic partial-volume edges

- Hard voxelisation gives aliased, staircase fibre edges — unlike CT.
- Supersampling averages many sub-voxel samples per voxel.
- Edge voxels take fractional values: the partial-volume effect.
- Here: 0 partial-volume voxels (hard) → 40 (supersampled $\times 5$).

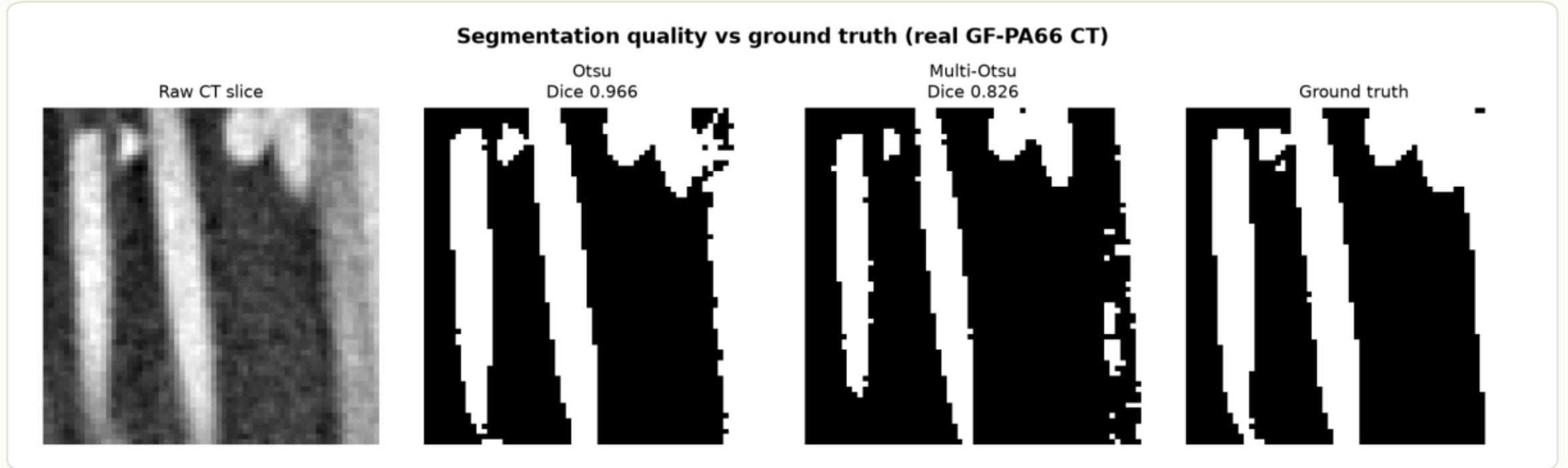


The resolved-regime pipeline



- Every stage is validated against ground truth.
- Segmentation method is switchable; its Dice is reported.
- Skeletonisation yields per-fibre length, tortuosity, orientation.

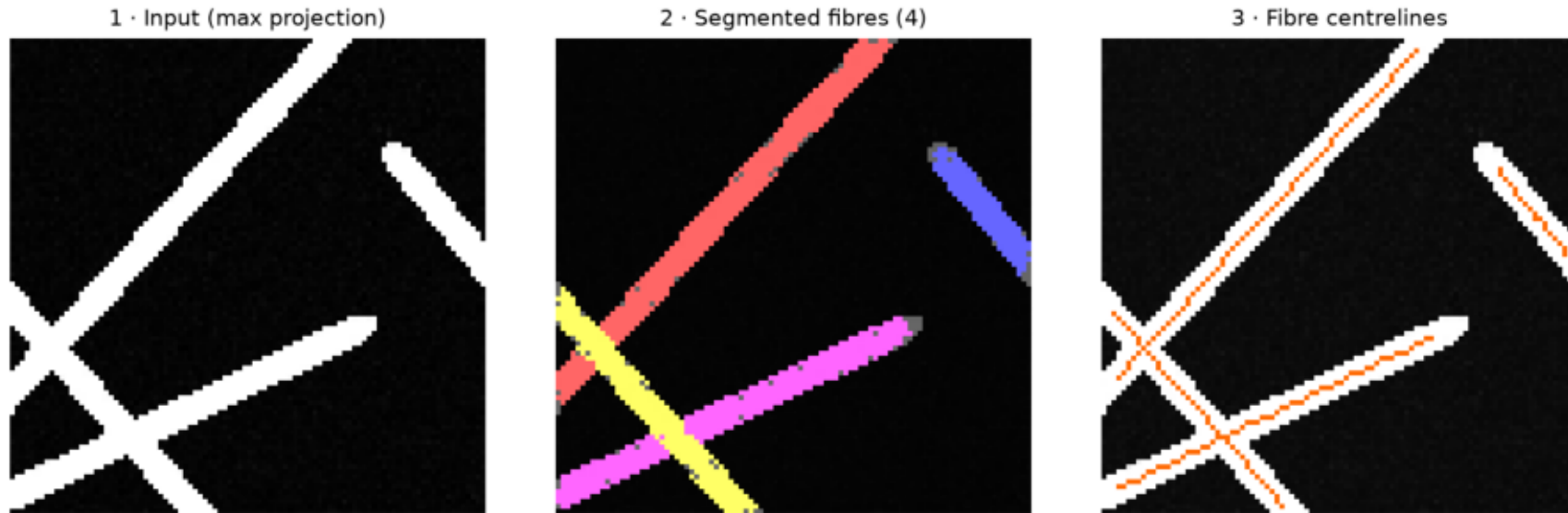
Validated segmentation on real CT



Real GF-PA66 X-ray CT vs ground-truth mask — Otsu reaches Dice 0.966. Method is switchable and its accuracy is reported per dataset.

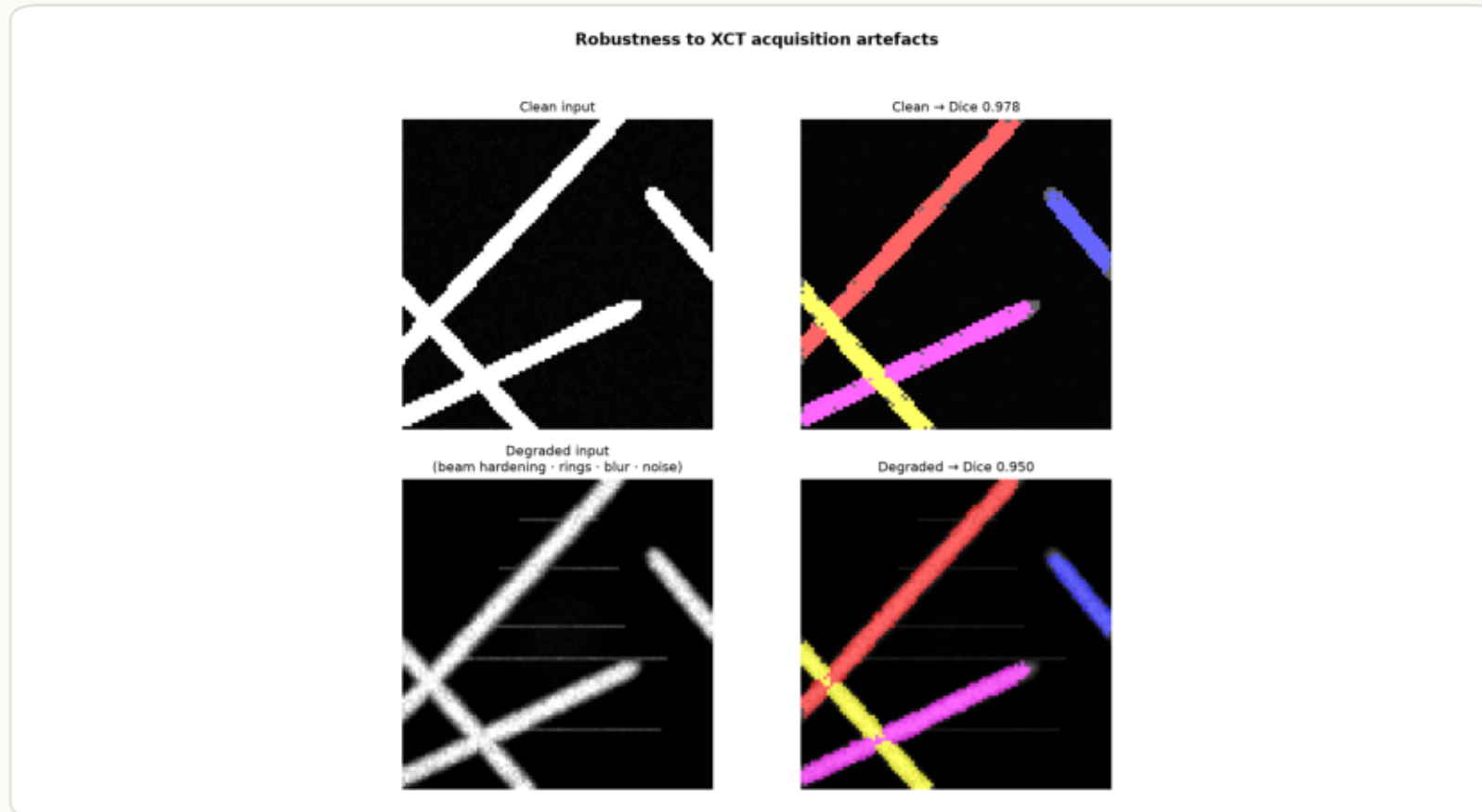
End-to-end: raw CT to fibre centrelines

Resolved pipeline · 4 fibres · Dice 0.978 · orientation error 0.20°



Four fibres segmented, labelled, skeletonised and oriented — mean orientation error 0.20°, Dice 0.978 vs ground truth.

Robust to acquisition artefacts



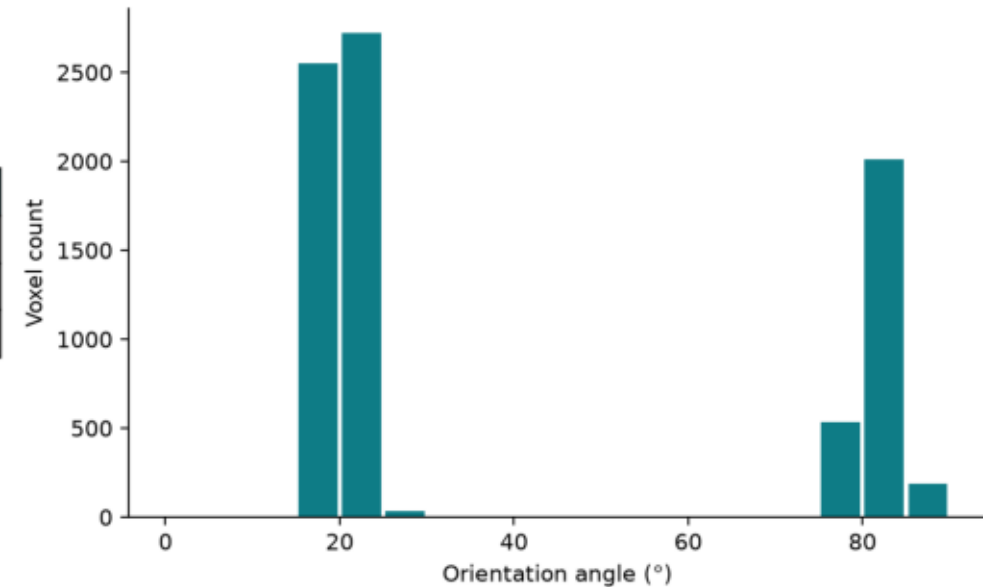
Strong beam hardening, rings, blur and noise added — segmentation Dice holds 0.978 → 0.950.

Regime awareness end to end

Auto-selected algorithm per resolution regime

Regime	Dice	Ang. err (°)	FA	Windows
resolved	0.978	0.20	-	-
marginal	-	-	0.650	729,000
subvoxel	-	-	0.618	-

Subvoxel orientation distribution



The same material at different voxel sizes: per-fibre morphometry when resolved, orientation-tensor fields (A2, FA) when marginal or subvoxel.

Validation summary

- Segmentation vs ground truth (real GF-PA66 CT): Dice 0.966.
- Resolved pipeline vs phantom: Dice 0.978, error 0.20°.
- Robustness to XCT artefacts: Dice 0.978 → 0.950.
- Phantom benchmark asserts Dice > 0.85 and angular error < 5°.
- Deterministic, seeded, one-command reproducible.

Takeaways & next steps

- Resolution-aware analysis gives defensible results across scan scales.
- Every number is validated against ground truth and reproducible.
- Next: learned 3D U-Net segmentation for low-contrast scans.
- Then: DVC and digital-twin coupling for prediction.

Key references

- Advani & Tucker (1987), J. Rheology — orientation tensors.
- Jeppesen et al. (2021), Composites A — structure-tensor orientation.
- Bertoldo et al. (2021), Frontiers in Materials — GF-PA66 benchmark.
- van der Walt et al. (2014), PeerJ — scikit-image.